Fadhil Muhammed N C

AI / Machine Learning Engineer

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Professional Summary

AI/ML engineer with hands-on experience in LLM agents, RAG, and tool calling, holding an MSc in Computer Science (Data Analytics, 2026) and a recent agentic AI internship. Built production features for a multi-tenant voice agent, including a usage-metering billing engine and an MCP server exposing booking, knowledge-base, and CRM tools, and cut meeting-assistant response latency about 2.4x. Also built ICU risk-prediction models over 61,000+ stays and a LangGraph multi-agent system deployed as four services on AWS ECS. Available to start immediately.

Technical Skills

Languages: Python, SQL
Agents and LLMs: LangGraph, MCP, RAG, FAISS, cross-encoder reranking, prompt engineering, QLoRA, Hugging Face
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Pandas, NumPy, YOLOv8, OpenCV
Backend and Cloud: FastAPI, Docker, AWS (ECS, ECR), GitHub Actions, MLflow

Experience

AI Intern - Agentic AI Developer Jul 2026 – Aug 2026
DifferentByte (ByteFlow), Kochi, India

- Built a wallet-based billing engine for a multi-tenant voice agent that meters per-session LLM, STT, TTS, and avatar usage and burns credits in real time, with booking, knowledge-base, and CRM tools exposed through a native MCP server.
- Implemented self-service multi-tenancy, a booking flow with CRM lead capture, and a reranked RAG knowledge base for a white-label chatbot live on a client website.
- Extended a meeting assistant across Google Meet, Zoom, and Teams with calendar-based scheduling for unattended joins.
- Migrated the assistant's LLM to Cerebras and tuned reasoning settings, cutting measured response latency about 2.4x without observed quality loss; began building an unattended, auto-scored HR interview mode.

ML Research Project — Smart ICU Assistant Jan 2026 – May 2026
Digital University Kerala | MIMIC-III | GitHub

- Built an ICU risk-prediction pipeline over 43.6 GB of MIMIC-III (61,532 stays), engineering 81 features per timestep and training LSTM, Transformer, XGBoost, and LightGBM models on chronological splits for 6 clinical tasks.
- Led sepsis, vasopressor, and ventilation prediction; a 4-model ensemble per task (AUROC-squared weighting, F1 threshold tuning) reached AUROCs of 0.77, 0.81, and 0.85, up to 0.03 above the best single model.
- Delivered a FastAPI dashboard with SHAP explanations and 8 rule-based alert types to triage patients by risk level.

Projects

Agentic Resume Intelligence System GitHub

- Built a 4-agent LangGraph pipeline (parser, JD analyst, gap analyst, rewriter) with conditional routing, skipping the rewriter on matches scoring below 30 to save 1 of 4 LLM calls, returning full analyses in 7–15 s.
- Implemented dual retrieval over a per-request FAISS index and 4 prebuilt corpora of 29 curated hiring documents, removing startup indexing and grounding every agent in domain context.
- Deployed 4 ECS Fargate services through a GitHub Actions pipeline, enabling push-to-deploy releases.

Finance Text-to-SQL Assistant GitHub

- Fine-tuned TinyLlama 1.1B with QLoRA (4-bit NF4, rank 16; 2.25M trainable parameters, 3 epochs, 4 GB VRAM) on 245 synthetic samples spanning 8 SQL pattern families over a six-table schema.
- Built an execution-based evaluator classifying syntax, hallucination, and logic errors; achieved 88.5% execution accuracy (23/26) on a small prompt-grouped holdout, with all 3 failures being logic errors.

Aerial Object Detection System GitHub

- Fine-tuned YOLOv8 for real-time aerial detection across a 10-class taxonomy, unifying VisDrone and UAVDT (22,584 images) into one YOLO-format dataset with automated class remapping.
- Exported YOLOv8n to ONNX with a from-scratch post-processing pipeline, improving CPU inference throughput 1.50× over PyTorch (8.21 vs. 5.47 FPS).

Education

MSc Computer Science, Data Analytics, Digital University Kerala (SGPA: 8.2) 2024 – 2026
BSc Mathematics, Farook College (Autonomous), Kozhikode 2021 – 2024

Certifications

Google Cloud Data Analytics Certificate May 2025
Google Cloud Computing Foundations Certificate July 2025